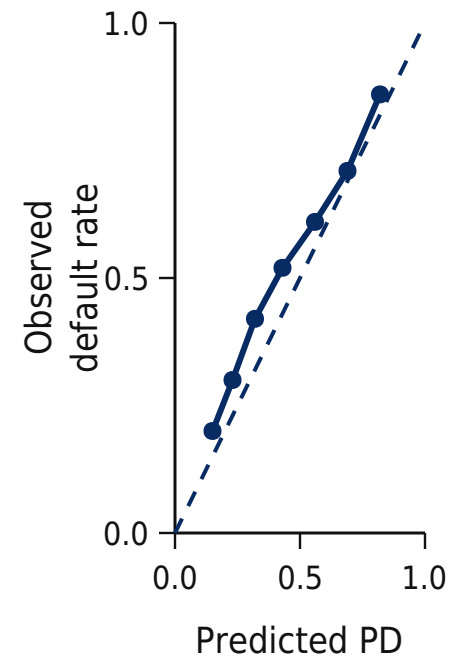
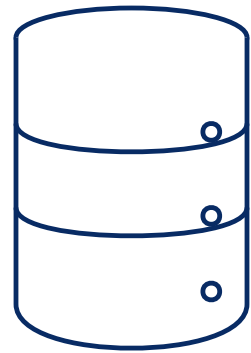


1 Frozen calibrated PD

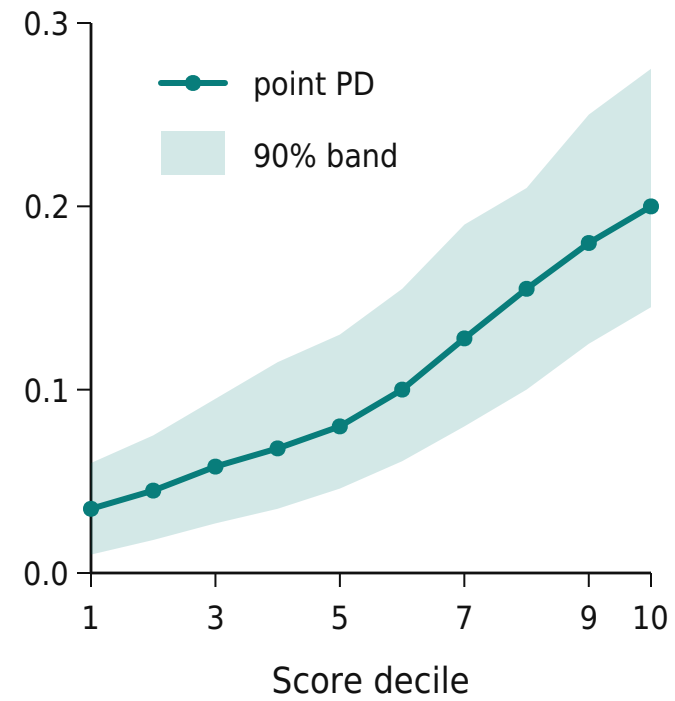


Lending Club OOT: n = 276,869

CatBoost + Venn-Abers

AUC 0.712, ECE 0.006

2 Mondrian conformal intervals



score-decile partition

90% coverage: 0.9297

min-group coverage: 0.9190

3 Robust portfolio optimization

$$\begin{aligned} &\max \sum_i x_i r_i \\ &\text{s.t.} \\ &\quad \text{budget} \\ &\quad \text{robust PD cap} \\ &\quad x_i \geq 0 \end{aligned}$$

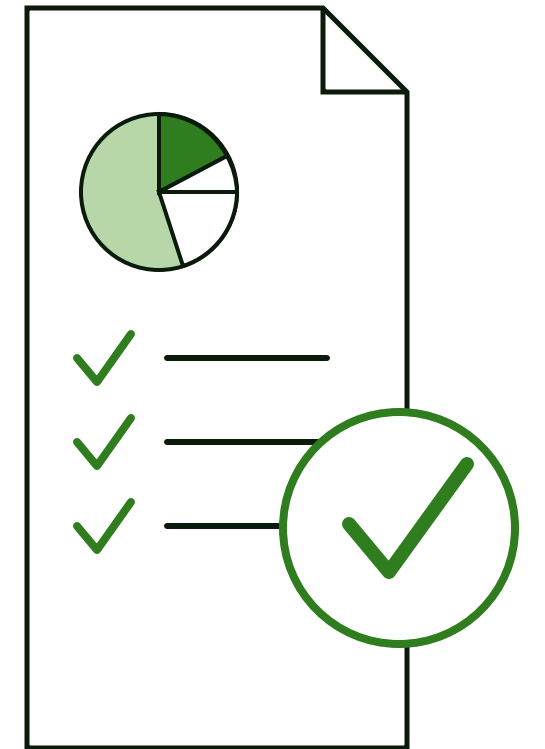


\$1M budget

point PD + conformal upper risk

$\tau = 0.175, \gamma = 0.45$

4 Promoted auditable policy



robust return: \$170.5K

$V(\alpha = 0.01) = 0.03645$

$\Gamma_{CP} = 0.18591$

exact pass; 45/45 safe policies

Prediction uncertainty becomes an auditable robust portfolio constraint.